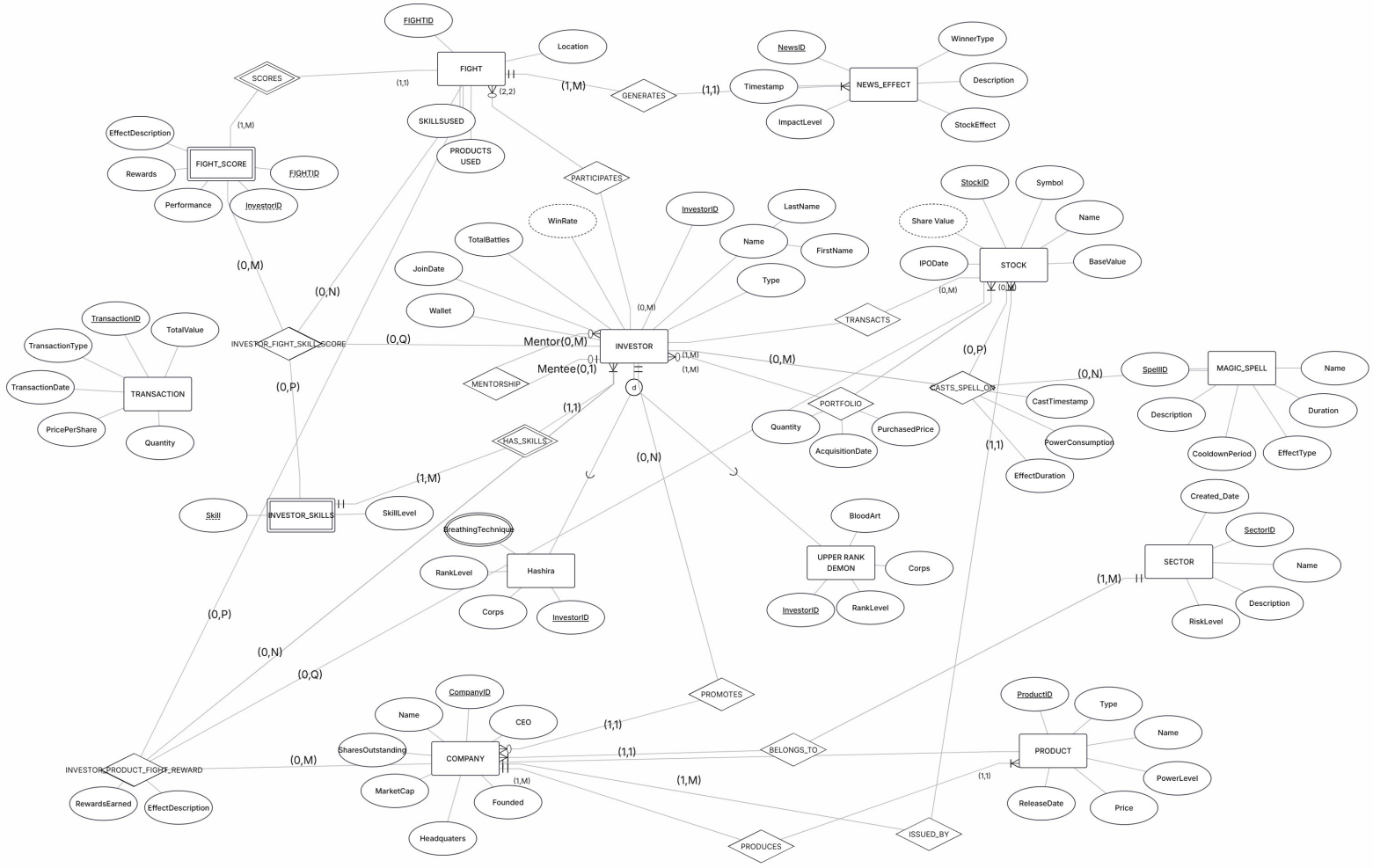

FIGHTID
Location
FIGHT
SCORES
(1,1)
(2,2)
(1,M)
GENERATES
(1,1)
NewsID
Timestamp
NEWS_EFFECT
WinnerType
Description
ImpactLevel
StockEffect
SKILLSUSED
PRODUCTS USED
EffectDescription
Rewards
FIGHT_SCORE
FIGHTID
Performance
InvestorID
(1,M)
PARTICIPATES
(0,M)
(0,N)
InvestorID
LastName
WinRate
TotalBattles
Name
FirstName
JoinDate
Type
Wallet
(0,M)
StockID
Symbol
Share Value
Name
IPODate
STOCK
BaseValue
(0,M)
(0,M)
TRANSACTS
TransactionID
TotalValue
TransactionType
INVESTOR_FIGHT_SKILL_SCORE
(0,Q)
Mentor(0,M)
INVESTOR
(1,M)
(1,M)
Mentee(0,1)
MENTORSHIP
(0,M)
(0,P)
CASTS_SPELL_ON
(0,N)
SpellID
MAGIC_SPELL
Name
TransactionDate
TRANSACTION
(0,P)
(1,1)
PORTFOLIO
CastTimestamp
Description
Duration
PricePerShare
Quantity
Quantity
PurchasedPrice
HAS_SKILLS
(0,N)
AcquisitionDate
PowerConsumption
(1,1)
CooldownPeriod
EffectType
EffectDuration
(1,M)
Created_Date
Skill
INVESTOR_SKILLS
SkillLevel
BreathingTechnique
BloodArt
SectorID
SECTOR
(1,M)
Name
UPPER RANK DEMON
Corps
RankLevel
Hashira
Description
Corps
InvestorID
InvestorID
RankLevel
RiskLevel
(0,P)
(0,N)
(0,Q)
PROMOTES
CompanyID
Name
CEO
(1,1)
SharesOutstanding
COMPANY
(1,1)
BELONGS_TO
ProductID
Type
Name
INVESTOR_PRODUCT_FIGHT_REWARD
(0,M)
(1,M)
PRODUCT
PowerLevel
MarketCap
Founded
(1,M)
(1,1)
RewardsEarned
EffectDescription
ReleaseDate
Price
Headquaters
PRODUCES
ISSUED_BY

CARBONATED-CAFFEINE PHASE-2

Minor Edits from Phase 1: No significant changes were made to the original requirements. All entities, relationships, attributes, and constraints from Phase 1 have been faithfully represented in the ER diagram using standard notation and ERDPlus tool capabilities.

5.1 Redundancy Analysis

The Hashira Market design minimizes redundancy through normalization, but a few potential overlaps exist.

Investor name data (FirstName + LastName and the derived InvestorName) may be repeated in related entities such as FIGHT, TRANSACTION, and PROMOTES when textual reports or logs store the full name instead of using InvestorID.

Problem: Repetition can cause inconsistencies if an investor changes their name, as one record may be updated while another remains old.

Company information (Company Name and CEO) might be redundantly stored in both COMPANY and STOCK tables if the stock record repeats descriptive fields for reporting.

Problem: If a company changes its CEO, every linked STOCK entry would also need updating.

5.2 Update Anomaly

Because of entity interconnections, a single update may need to propagate across several tables.

Example: Changing a company's MarketCap in COMPANY must also update derived stock values in STOCK and PORTFOLIO tables where investors hold that company's shares.

Problem: If these updates occur only in one table, portfolio valuations and reports may display inconsistent figures.

5.3 Insertion Anomaly

Some entities cannot be inserted without existing related data.

Scenario: A PRODUCT record cannot be inserted until a valid CompanyID exists in COMPANY.

Problem: If a new company is still under registration, its products cannot be recorded even if their details are known. Similarly, inserting a FIGHT requires pre-existing INVESTOR records for both the winner and the loser.

5.4 Deletion Anomaly

Deleting parent records could unintentionally remove important dependent information.

Example: Deleting a COMPANY causes cascading deletion of PRODUCT, STOCK, and PORTFOLIO entries, which may erase critical transaction history.

Problem: Historical fight or investment data tied to the deleted company would vanish, resulting in the loss of valuable audit and trend information.

Another case is deleting a FIGHT, which removes associated NEWS_EFFECT entries and results in the loss of evidence about market impact events.